

Customer Segmentation and Review Score Prediction in Olist E-Commerce

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1 Executive Summary

This project analyzed the Olist Brazilian E-Commerce dataset to understand customer behavior and predict satisfaction. Using RFM analysis and K-Means clustering, customers were segmented into distinct groups, revealing that high-value buyers drive disproportionate revenue while many customers are at risk of churn. Logistic Regression, Random Forest, and Gradient Boosting models were tested to predict review scores, with Gradient Boosting performing best. The results show that delivery delays are the strongest driver of dissatisfaction, payment type influences satisfaction, and loyal customers require targeted retention. I recommend Olist invest in delivery reliability, create retention programs for high-value clusters, and optimize payment offerings to improve customer experience and lifetime value.

2 Problem Statement

Olist, Brazil's largest online marketplace, connects thousands of sellers with customers across the country. While the platform has experienced rapid growth, it faces the ongoing challenge of ensuring consistent customer satisfaction and building long-term loyalty in a highly competitive e-commerce environment. Negative experiences, particularly delivery delays and lapses in service quality, can lead to low review scores, fewer repeat purchases, and ultimately customer churn.

This project addresses the question of how Olist can better understand customer behavior and predict satisfaction in order to improve retention and lifetime value. To approach this problem, the analysis focused on two complementary objectives. The first was customer segmentation: grouping customers into meaningful categories using Recency, Frequency, and Monetary (RFM) patterns to identify loyal buyers, one-time purchasers, and at-risk customers. The second was satisfaction prediction: building predictive models to uncover the factors most strongly associated with customer review scores, which can guide proactive interventions to enhance the customer experience.

By tackling these objectives, the project provides Olist with a framework to prioritize operational improvements, design targeted retention strategies, and ultimately increase revenue by focusing on the customers and processes that matter most.

3 Data Overview

This project uses the Olist Brazilian E-Commerce Public Dataset from Kaggle, which consolidates transactional data from over 100,000 orders placed between 2016 and 2018 across multiple marketplaces in Brazil. The dataset is structured into multiple linked tables, each capturing a different aspect of the e-commerce process:

- **Customers:** Customer IDs, location, and unique identifiers.
- **Orders:** Order status, timestamps for purchase, approval, shipping, and delivery.
- **Order Items:** Product details, seller IDs, prices, and freight values.
- **Products:** Product IDs, category names, and physical attributes including dimensions and weight.
- **Payments:** Payment methods, number of installments, and payment values.
- **Reviews:** Customer review scores and free-text feedback.

- **Sellers:** Seller IDs and locations.
- **Geolocation:** Latitude and longitude data for customers and sellers.
- **Product Category Translations:** Mapping from Portuguese to English product categories.

For this analysis, the tables were merged into an integrated analytical dataset at the order level, enabling exploration of customer behavior, satisfaction, and financial metrics.

After cleaning and preprocessing (handling missing values, aligning timestamps, and standardizing categorical labels), the dataset provided a robust foundation for exploratory analysis, customer segmentation, and predictive modeling.

4 Data Wrangling and Cleaning

The Olist dataset consists of multiple relational tables capturing different aspects of the e-commerce process, including customers, orders, items, products, payments, sellers, and reviews. To prepare these data for analysis, several wrangling and cleaning steps were performed to create a consistent, analysis-ready foundation.

The first step was integrating the data sources. Core tables—such as customers, orders, order items, payments, reviews, and sellers—were merged using unique identifiers like `order_id` and `customer_id`. Product category translations were joined to replace Portuguese names with English equivalents, improving interpretability, while geolocation data were incorporated to align customer and seller locations where relevant.

Cleaning procedures addressed data quality issues across the merged dataset. All timestamp variables (purchase, approval, shipping, delivery, and review creation dates) were converted into datetime objects to allow for accurate time-based calculations. Records with missing values in critical fields, such as delivery dates or review scores, were removed, while non-critical nulls (e.g., missing geolocations) were retained, as they did not affect downstream analysis. Duplicate records were checked and eliminated to preserve the uniqueness of both `order_id` and `customer_id`. Outlier detection was also conducted, and implausible values such as negative delivery times or zero-priced orders were excluded.

Through these steps, the raw Olist dataset was transformed into a clean and integrated analytical dataset. This provided a solid foundation for the subsequent stages of exploratory data analysis, feature engineering, and modeling.

5 Exploratory Data Analysis

Exploratory data analysis was conducted to uncover patterns in customer satisfaction, order characteristics, and purchasing behavior, and to guide the development of both clustering and predictive models. The analysis focused on review scores, delivery performance, product characteristics, payment methods, and Recency–Frequency–Monetary (RFM) patterns.

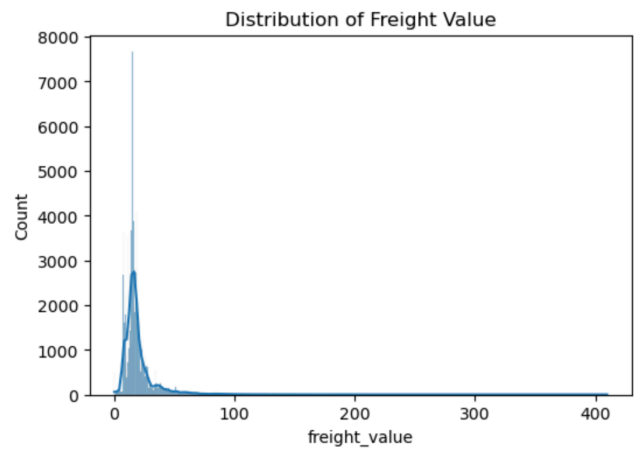
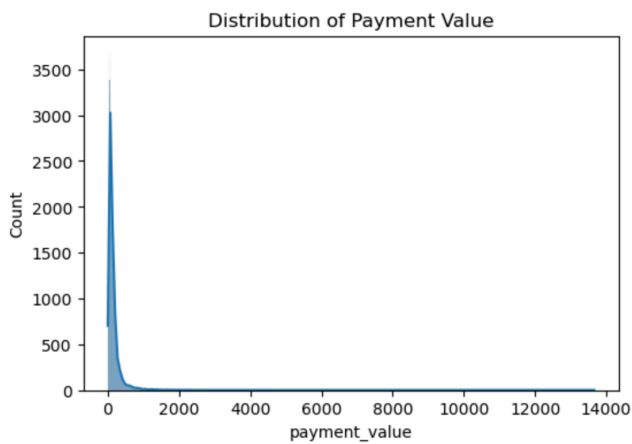
5.1 Univariate Analysis

Univariate analysis provided an initial view of the key variables. Review scores were highly skewed toward five stars, with the majority of customers reporting very positive experiences.

However, a meaningful portion of one- and two-star reviews highlighted the presence of dissatisfied customers whose concerns warranted deeper investigation.



Order values and freight charges were also right-skewed, with most purchases falling at lower price points and only a small number of high-value transactions contributing disproportionately to overall revenue.

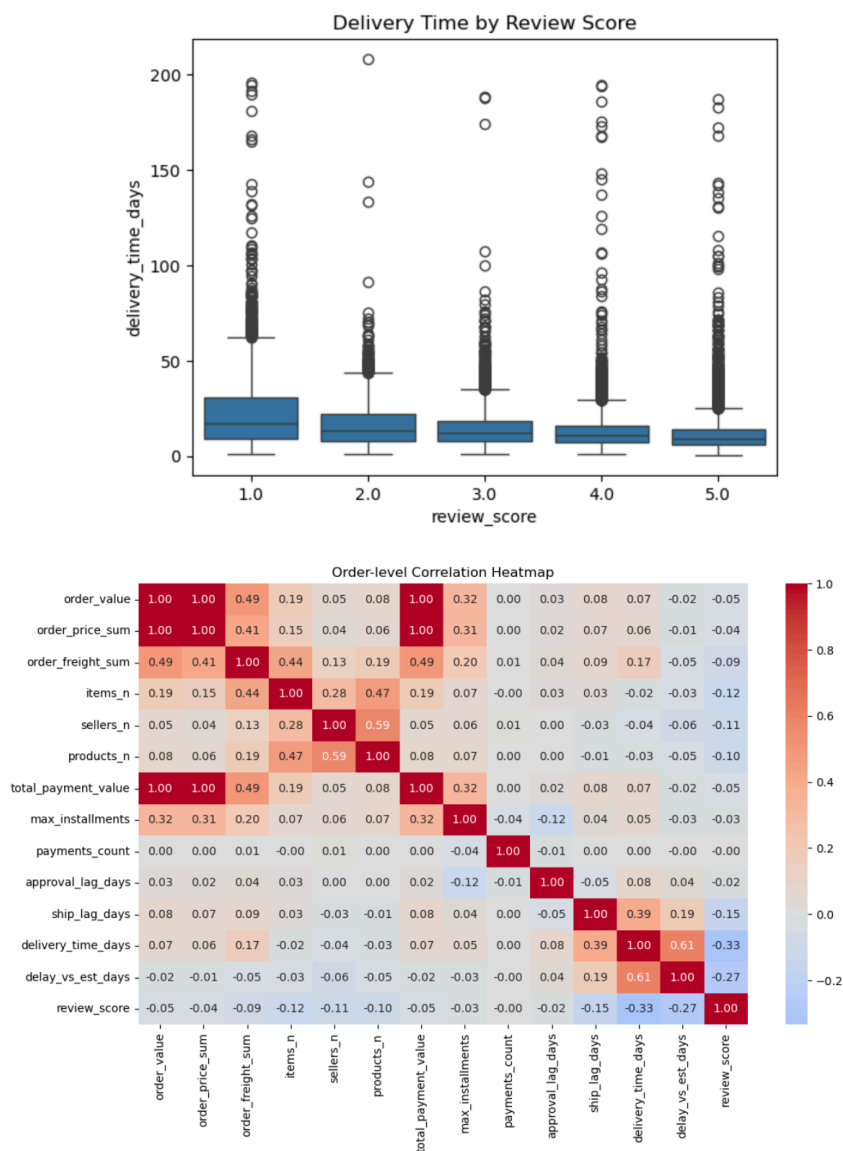


5.2 Bivariate Analysis

Bivariate analysis was used to explore relationships between customer satisfaction and key operational or order-related features. This step provided insight into which factors most strongly influenced review scores and guided feature selection for predictive modeling.

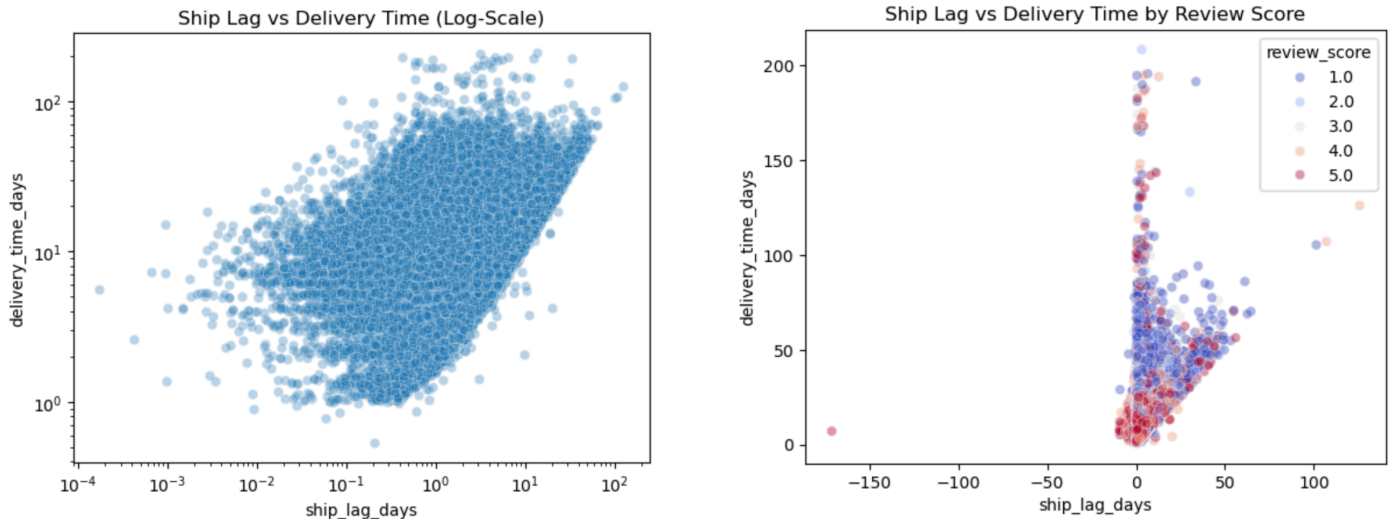
5.3 Review Scores and Satisfaction

Customer reviews served as the primary proxy for satisfaction. The distribution of review scores was heavily skewed toward five stars, suggesting that most customers were satisfied with their purchases. However, low review scores were still prevalent enough to warrant attention, particularly when examined alongside operational features. A boxplot of delivery delay against review scores revealed that longer delivery delays consistently led to lower ratings. Correlation analysis further confirmed that delivery performance had the strongest relationship with customer satisfaction.



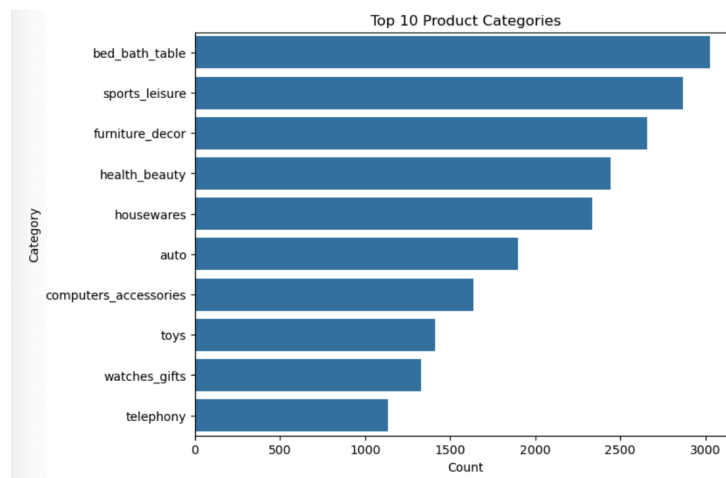
5.4 Delivery Performance

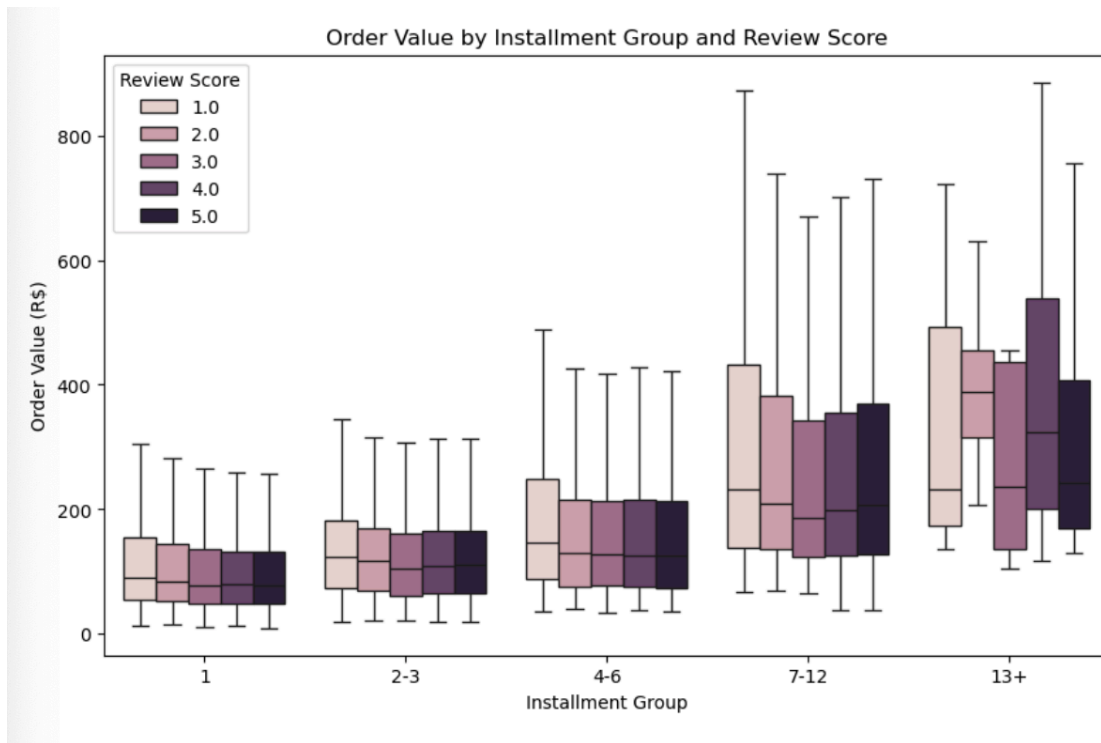
Examining delivery times provided insight into Olist's logistics operations. Actual delivery dates often exceeded the estimated delivery dates, indicating that delays were common. A scatterplot of delivery delays versus review scores reinforced the earlier finding: as delays increased, ratings declined sharply. Conversely, on-time and early deliveries were strongly associated with positive reviews, making delivery reliability a clear target for improvement.



5.5 Order Characteristics

Order characteristics also provided useful context. The most common product categories included home goods, sports and leisure, furniture, and health and beauty. More telling were differences by order value: higher-value orders showed greater variability in satisfaction, with both exceptionally positive and negative reviews. This indicates that when customers spend more, their expectations may also be higher, amplifying both positive and negative experiences.





6 Feature Engineering and Preprocessing

Following exploratory analysis, several new features were engineered to capture customer behavior and operational performance more effectively. These features were designed to strengthen both the clustering analysis and predictive modeling tasks.

At the customer level, Recency, Frequency, and Monetary (RFM) metrics were created to summarize purchasing behavior. Recency was defined as the number of days since a customer's most recent review, Frequency as the total number of completed orders, and Monetary as the total revenue generated by each customer. These measures captured different dimensions of customer engagement and purchasing value. Because their distributions were highly skewed, the RFM features were standardized prior to use in clustering and modeling.

Operational features were also engineered from the order-level data. Delivery delay was calculated as the difference between the estimated delivery date and the actual delivery date, providing a quantitative measure of fulfillment performance. Order value was defined as the sum of product price and freight value, representing the total amount paid for each transaction. Payment method was encoded into dummy variables to allow categorical information, such as credit card, boleto, debit, and voucher, to be incorporated into machine learning models.

After feature engineering, preprocessing steps ensured that the dataset was suitable for modeling. Continuous features such as delivery delay, order value, and the RFM scores were standardized to place them on a comparable scale, which is particularly important for distance-based methods such as K-Means clustering and for regularized regression models. Categorical variables, including payment method, were one-hot encoded to allow them to be used effectively by machine learning algorithms. Finally, the dataset was split into training and test sets to evaluate model performance in a controlled manner.

Through this process, the raw transactional data was transformed into a rich, analysis-ready dataset that integrated behavioral, operational, and transactional dimensions. These engineered features provided the foundation for both customer segmentation and the predictive modeling of customer satisfaction.

7 Modeling

This project involved two complementary modeling approaches. The first was unsupervised clustering, designed to group customers into meaningful behavioral segments. The second was supervised classification, aimed at predicting customer satisfaction as measured by review scores. Together, these approaches provided insights into both who Olist's customers are and what drives their satisfaction.

7.1 Customer Segmentation with Clustering

Customer segmentation was performed using K-Means clustering on the standardized Recency, Frequency, and Monetary (RFM) features. The goal was to identify distinct customer groups based on purchasing behavior. To determine the appropriate number of clusters, both the elbow method and silhouette scores were applied. Using the elbow method, diagnostics suggested that a four-cluster solution provided a good balance between compactness and interpretability. Rule based RFM clustering was also performed to provide an alternative perspective. Ultimately, K-Means provided more nuanced and data-driven results.

The resulting clusters revealed clear behavioral patterns. One group represented high-frequency, high-monetary customers who were likely to be Olist's most valuable buyers. Another consisted of customers with infrequent and low-value purchases, representing the majority of Olist's base but with limited long-term value. Additional segments included customers who were recent but low-frequency purchasers, and those who were historically valuable but at risk of churn due to inactivity. These clusters provided actionable insight for targeted retention and marketing strategies. These groups were given the common clustering group names of Champion, Loyal, At Risk, and Lost.

7.2 Satisfaction Prediction with Supervised Learning

To complement the segmentation analysis, regression models were developed to predict customer review scores as a continuous outcome. Three models were selected for comparison: Linear Regression, Random Forest Regressor, and Gradient Boosting Regressor. Linear Regression provided a simple and interpretable baseline, Random Forest allowed for the modeling of nonlinear relationships and interaction effects, and Gradient Boosting was expected to achieve strong predictive performance by sequentially minimizing residual errors.

The dataset was split into training and test sets, with models trained on the training data and evaluated on the holdout test set. Performance was measured using regression metrics including R^2 , Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE).

Among the models, the Gradient Boosting Regressor achieved the best balance of predictive power and generalization.

8 Results

8.1 Customer Segmentation Results

The K-Means clustering analysis on standardized RFM features identified four distinct customer segments. These groups reflected meaningful differences in purchasing behavior:

Cluster 1: Loyal High-Value Buyers

Customers with frequent, recent purchases and high total spending. This segment represented a small percentage of the customer base but contributed disproportionately to total revenue.

Cluster 2: At-Risk Customers

Historically active customers whose recency scores indicated declining engagement.

Cluster 3: One-Time Low-Value Buyers

The largest segment, consisting of customers who made a single low-value purchase and showed no signs of repeat behavior.

Cluster 4: Recent Low-Value Buyers

Customers who purchased recently but had low frequency and monetary values, representing potential growth opportunities.

A cluster profile table can summarize the average recency, frequency, and monetary scores for each group. Visualizations such as RFM scatterplots or bar charts of cluster proportions help illustrate the long-tail distribution of Olist's customer base.

Segmentation revealed that a small minority of high-value customers drive most of Olist's revenue, while the majority are one-time buyers. This insight highlights the need for targeted retention strategies.

	n_customers	recency	frequency	monetary	sil_mean
cluster					
0	28103	163.197523	1.000000	320.699013	0.310430
3	2868	211.016736	2.114017	310.039554	0.648098
1	27148	417.513408	1.000000	119.465852	0.366011
2	36585	138.609621	1.000000	69.841908	0.405153



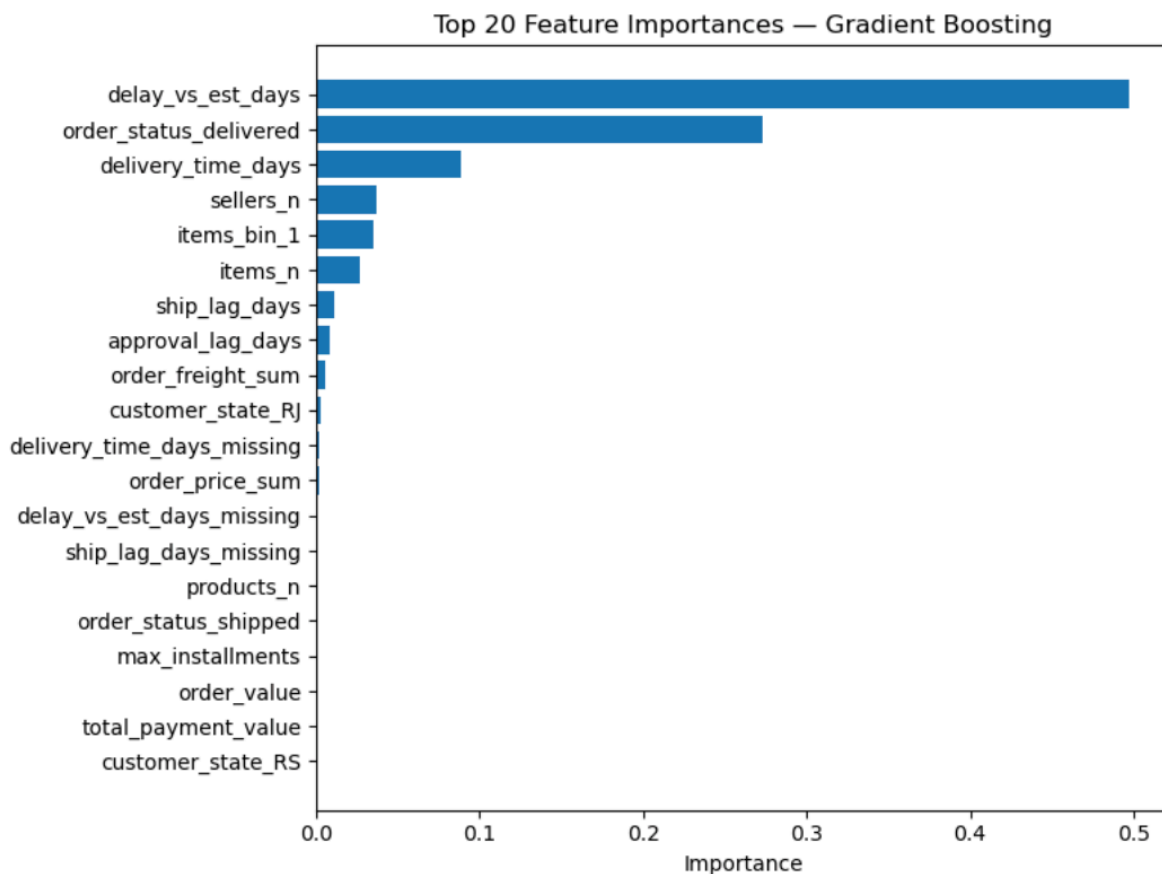
8.2 Satisfaction Prediction Results

Model	RMSE	MAE	R ²
Baseline (Mean)	1.324	1.035	0.000
Linear Regression	1.184	0.911	0.200
Random Forest	1.159	0.893	0.234
Gradient Boosting	1.139	0.873	0.260

The Gradient Boosting Regressor achieved the highest R² and the lowest RMSE, indicating that it captured more variance in review scores and produced more accurate predictions than the baseline linear model. Random Forest also performed well but slightly underperformed compared to Gradient Boosting. Linear Regression, while less accurate, remained useful for interpretability.

Feature importance analysis showed that delivery delay was the strongest predictor of review scores. This confirmed the findings from exploratory analysis and quantified their relative influence on satisfaction.

The results reinforced that operational efficiency in delivery has the largest impact on customer satisfaction, while monetary value and payment methods play secondary roles



9 Recommendations

The analyses from this project suggest several actions Olist can take to strengthen customer satisfaction and increase long-term value.

1. Improve delivery reliability and transparency.

Delivery delays consistently emerged as the strongest predictor of dissatisfaction. Olist should invest in operational improvements to reduce shipping and delivery times, such as better coordination with sellers, tighter logistics partnerships, and improved tracking systems. In the short term, even greater transparency in estimated delivery dates could help manage customer expectations and prevent negative experiences.

2. Prioritize high-value customer clusters.

Clustering revealed that a small proportion of loyal, high-frequency buyers contribute disproportionately to revenue. These customers should be the focus of retention strategies such as loyalty programs, personalized promotions, or proactive customer service. Protecting this group from churn will have an outsized impact on Olist's revenue.

3. Develop strategies for at-risk and new customers.

Segments with low frequency or declining recency represent opportunities for growth if they can be re-engaged. Olist could use targeted campaigns, such as discounts for second purchases or win-back offers for inactive accounts, to encourage repeat buying and move customers toward higher-value segments.

Together, these recommendations provide Olist with both strategic and operational levers. Improving delivery performance will address the root cause of dissatisfaction, while customer segmentation allows for tailored retention and growth strategies. Optimizing payment methods adds another layer of customer experience improvement. Taken together, these steps can help Olist improve satisfaction, reduce churn, and increase lifetime value.

10 Limitations and Future Work

This project demonstrates how customer segmentation and satisfaction prediction can provide actionable insights for Olist. However, there are several limitations that should be considered when interpreting the results.

One limitation lies in the dataset itself. The Olist dataset captures transactions between 2016 and 2018 and may not fully represent current e-commerce dynamics. Additionally, it does not include customer demographic information, marketing campaign exposure, or product-level details such as brand reputation or quality, all of which could influence satisfaction and loyalty.

Another limitation is related to modeling assumptions. The clustering analysis relied on K-Means, which requires the number of clusters to be specified in advance and assumes spherical cluster shapes. Alternative clustering methods, such as DBSCAN or hierarchical clustering, could capture more complex structures in the data. Similarly, the regression models explained a portion of the variance in review scores but left room for unexplained factors. Customer satisfaction is multifaceted, and not all drivers can be captured through transactional data alone. Additional hyper parameter tuning or the construction of ensemble model may provide a more accurate predictor of customer satisfaction.

Finally, there are practical considerations for implementation. While the models identified clear drivers of satisfaction, translating these into operational changes requires coordination across logistics, seller management, and customer service. Improvements in delivery performance, for example, may depend on external partners beyond Olist's direct control.

11 Conclusions

This project set out to help Olist, Brazil's largest online marketplace, better understand its customers and improve satisfaction in a highly competitive e-commerce environment. By combining customer segmentation and predictive modeling, the analysis provided both a descriptive view of Olist's customer base and a predictive understanding of what drives satisfaction.

Clustering on Recency, Frequency, and Monetary features revealed that most customers are one-time buyers with limited long-term value, while a small but important segment of loyal, high-spending customers contributes disproportionately to revenue. This insight underscores the need for tailored retention strategies that protect and grow Olist's most valuable relationships.

Regression modeling confirmed that delivery performance is the strongest driver of review outcomes. These results highlight the importance of operational reliability and customer experience in shaping perceptions of the marketplace.

Together, these analyses provide Olist with a data-driven framework to guide business decisions. By improving delivery reliability, optimizing payment options, and tailoring retention strategies to specific customer segments, Olist can reduce churn, increase satisfaction, and maximize lifetime value.

12 References

Olist Brazilian E-Commerce Public Dataset. Kaggle.

<https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce>